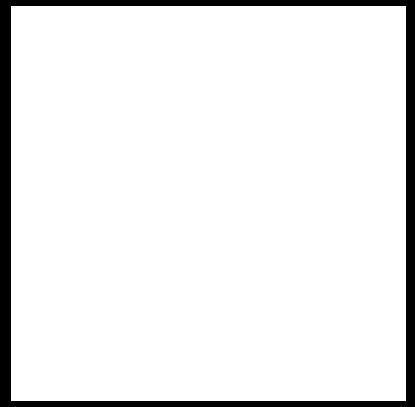
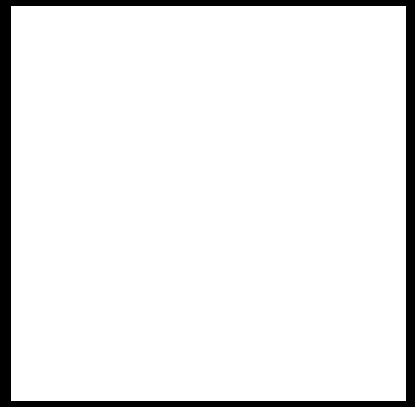
ual: creative computing institute

Coding for Digital Media
Week 4 - Cut and Paste Audio

Louis McCallum Autumn 2026



Physics of Sound



What is Sound?

- Sound is vibrations
- Can travel through air, or other matter

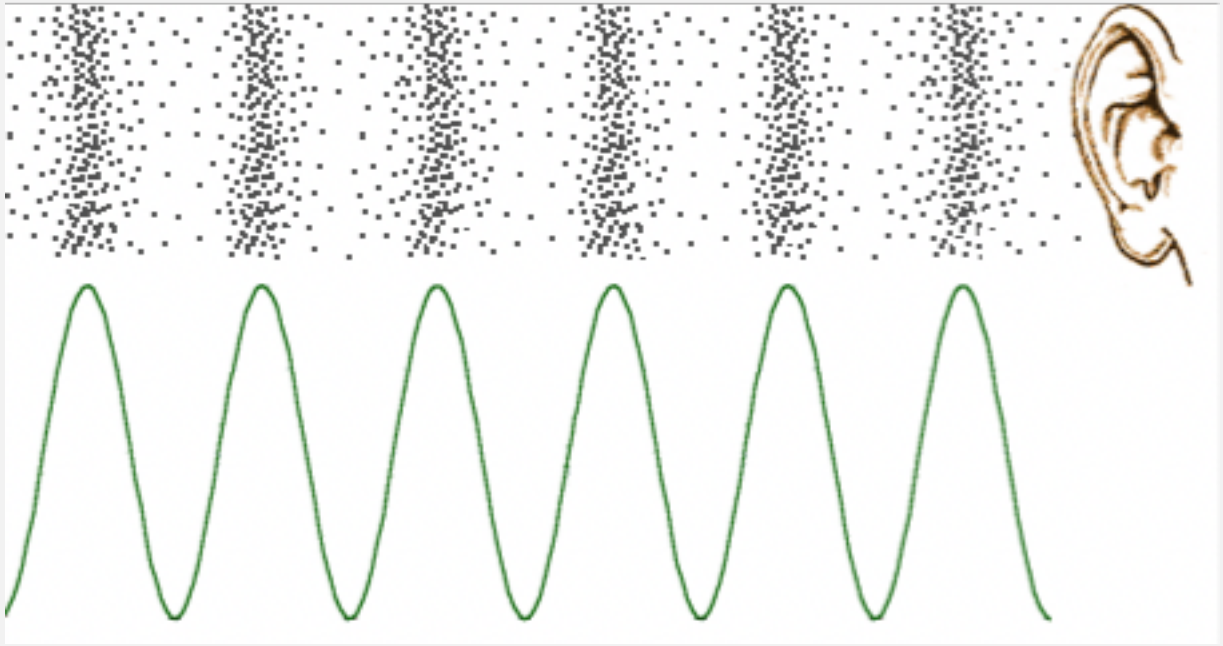


https://www.youtube.com/watch?v=lxsM9Jv0OD8

What is Sound?

- Air molecules compressing and rarefying





What is Sound?

- Air molecules compressing and rarefying

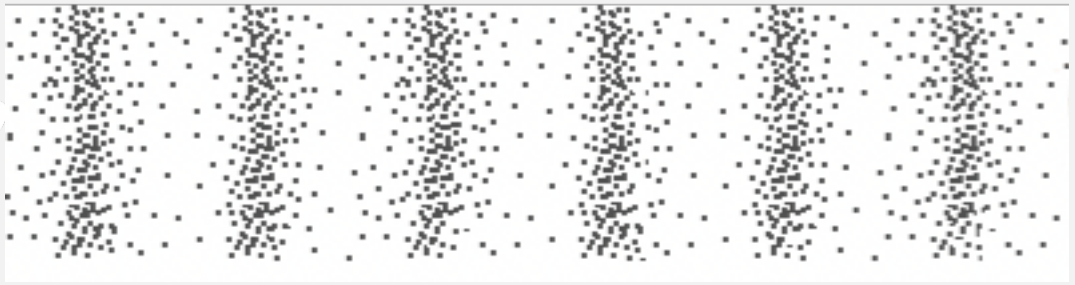


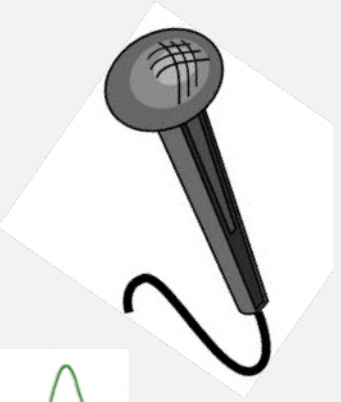
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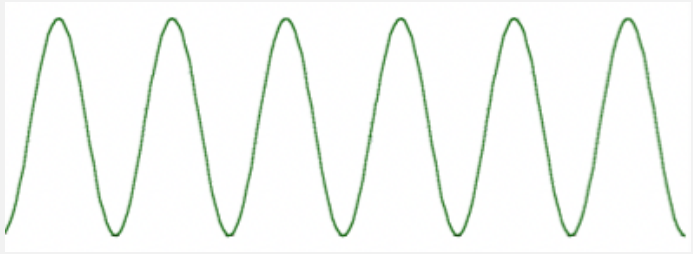
Recording Sound

- Vibrations in air turn to changes in voltage





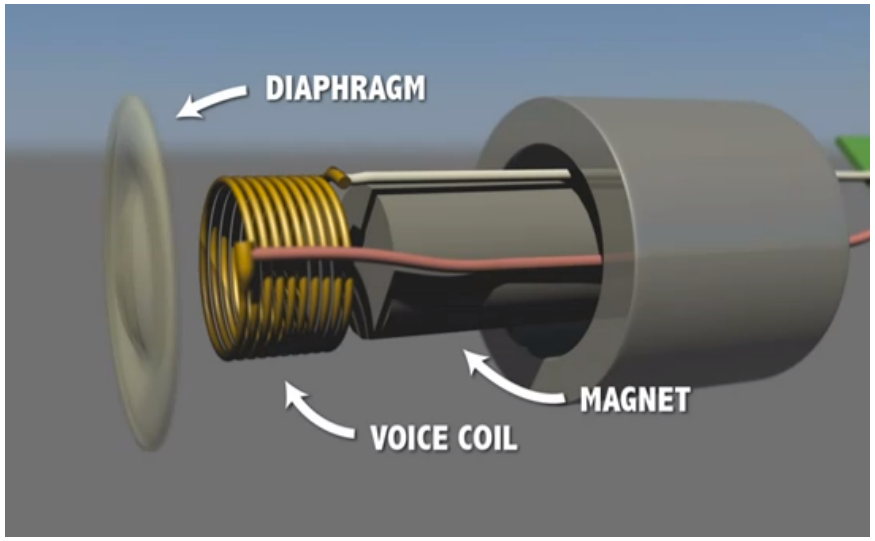




Recording Sound

- Vibrations in air turn to changes in voltage

Inside a dynamic microphone



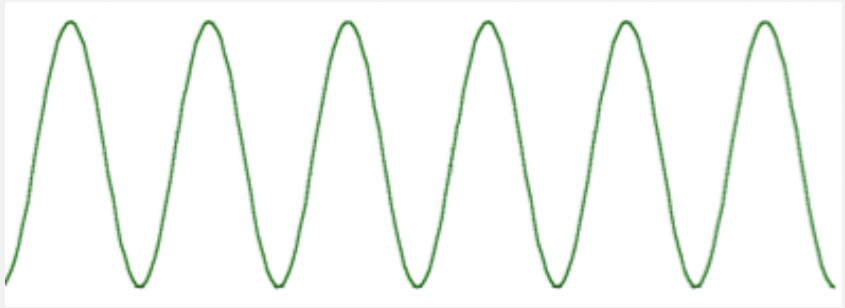
Mic is a **transducer**: Converts a signal from one form of energy into another

16

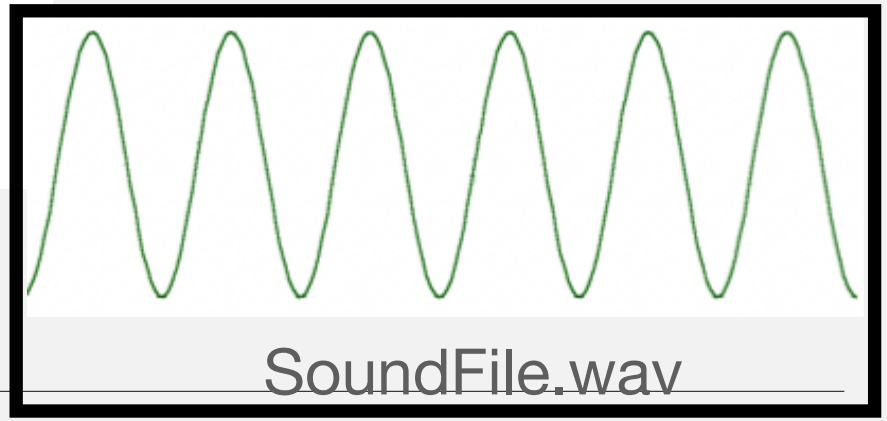
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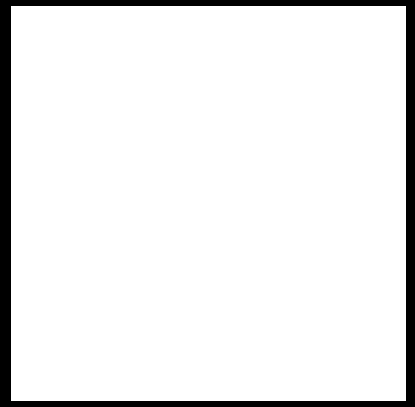
Recording Sound

- Analog Digital Converters (ADC)

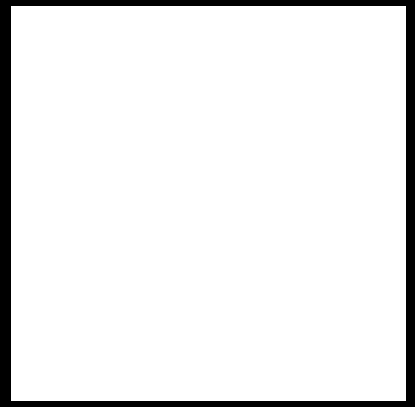






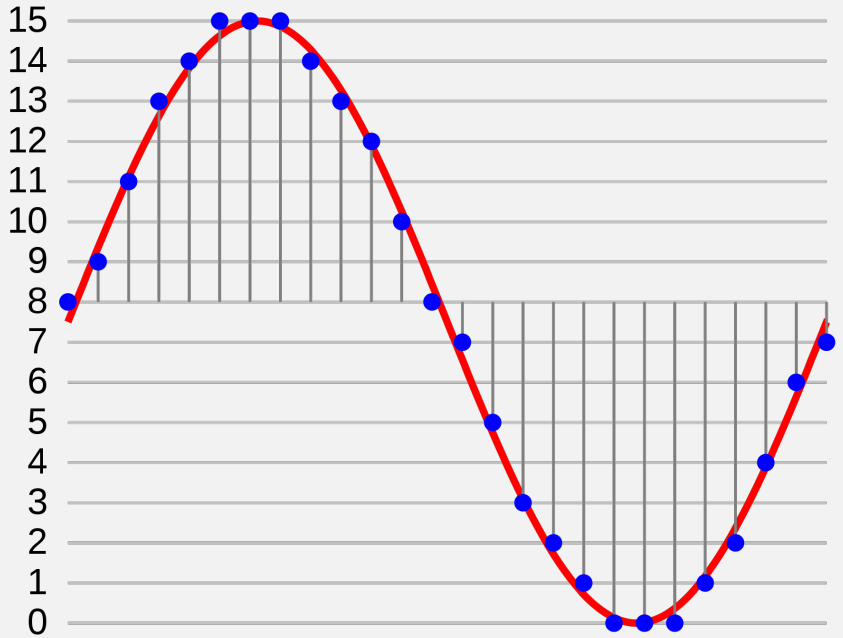


Digital Audio



Sampling

- We can't represent continuous signals on a computer
- So we sample it at regular intervals
- The amount of times we do this per second is the sample rate
- These numbers are what is in a uncompressed audio file e.g. .wav



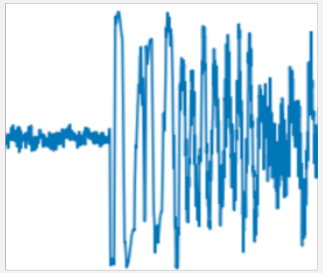
Audio Files

- The contents is just the samples that make up the waveform
- Can multiply all values to change volume
- Can change the speed and pitch by playing back samples faster or slower (given a constant sample rate

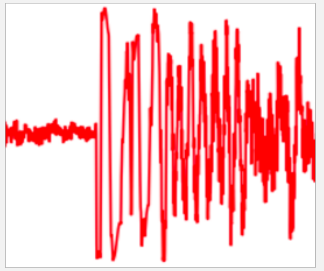
Audio Files

- Depending on the play back sample rate, you need different numbers of samples to make up the same the length of audio file

44100Hz
44100 samples = 1 second



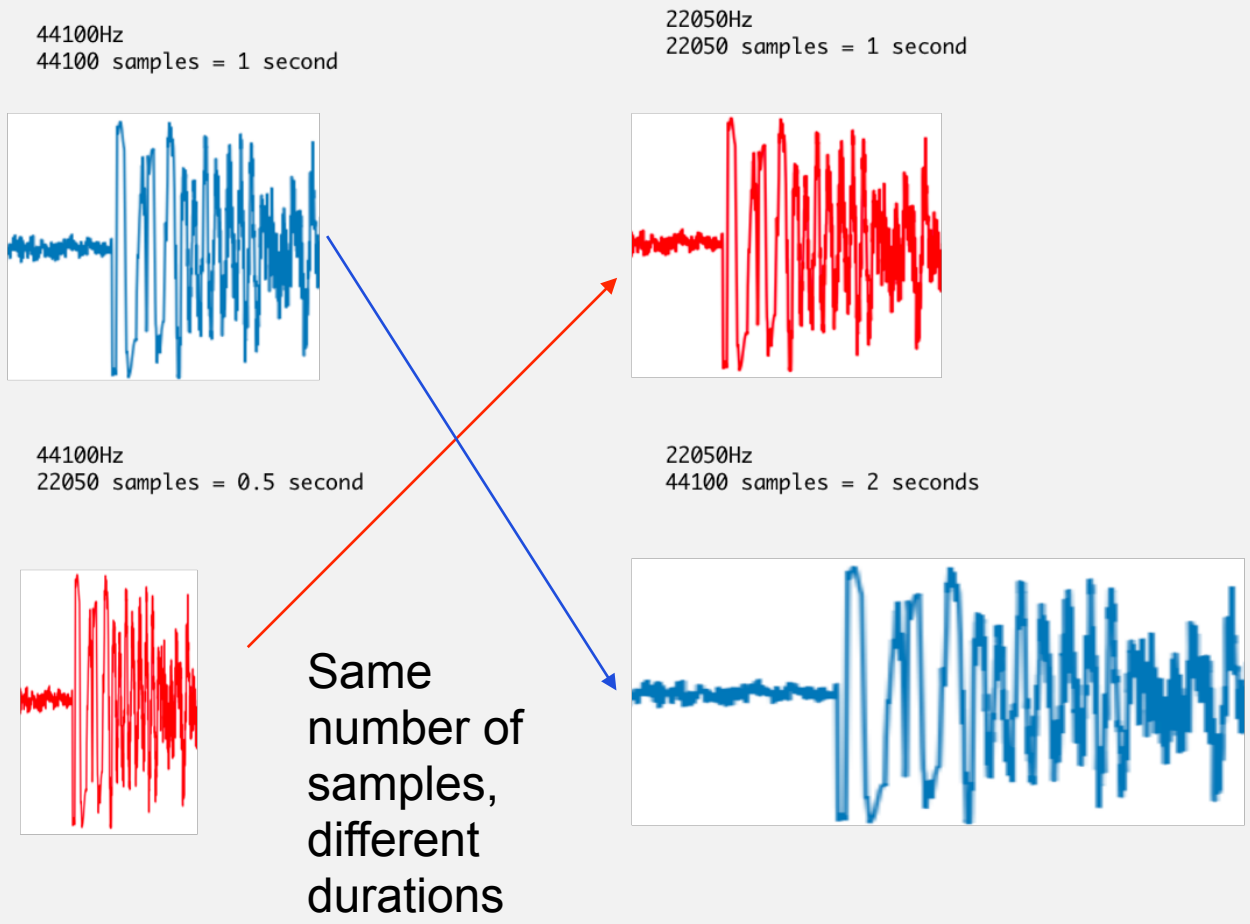
22050Hz
22050 samples = 1 second

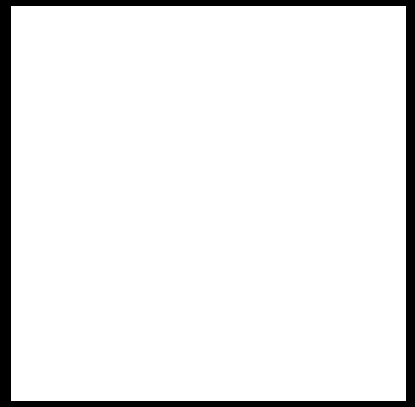


Both 1 second long!

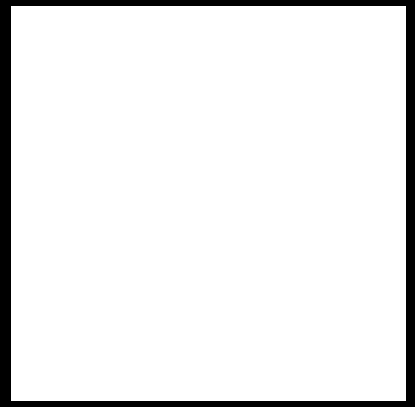
Audio Files

- Depending on the sample rate, the same number of samples can provide audio for different lengths of time





Sampling



Cutting Tape

6

One of the devices which the 20th Century offers the composer is the tape recorder, a remarkable piece of technology.

Composers have used the physical cut and paste of tape to make new compositions since the early 20th century



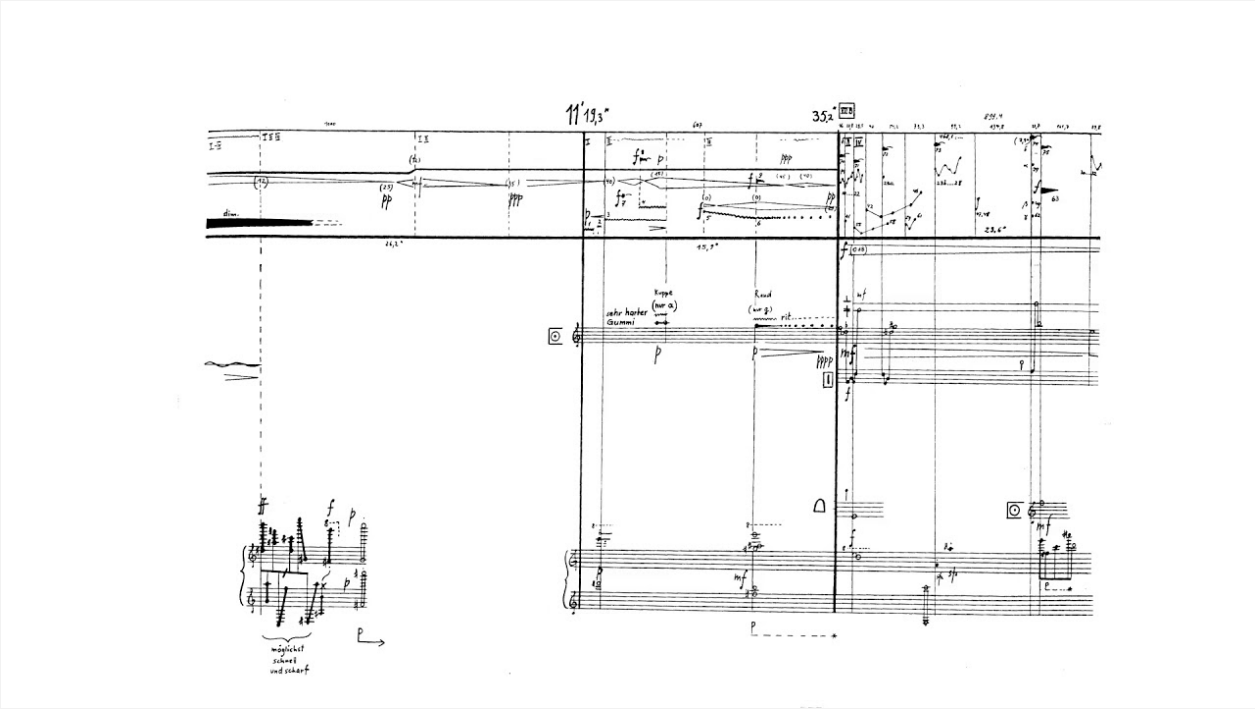
Early Cut and Paste

- Often credited with pioneering *musique concrète* in the 1940s in France, Pierre Schaeffer's used recorded sounds (instead of musical notation) as raw material
- His work involved cutting, splicing, and manipulating magnetic tape
- *Étude aux chemins de fer (1948)* is considered one of the first pieces of musique concrète.



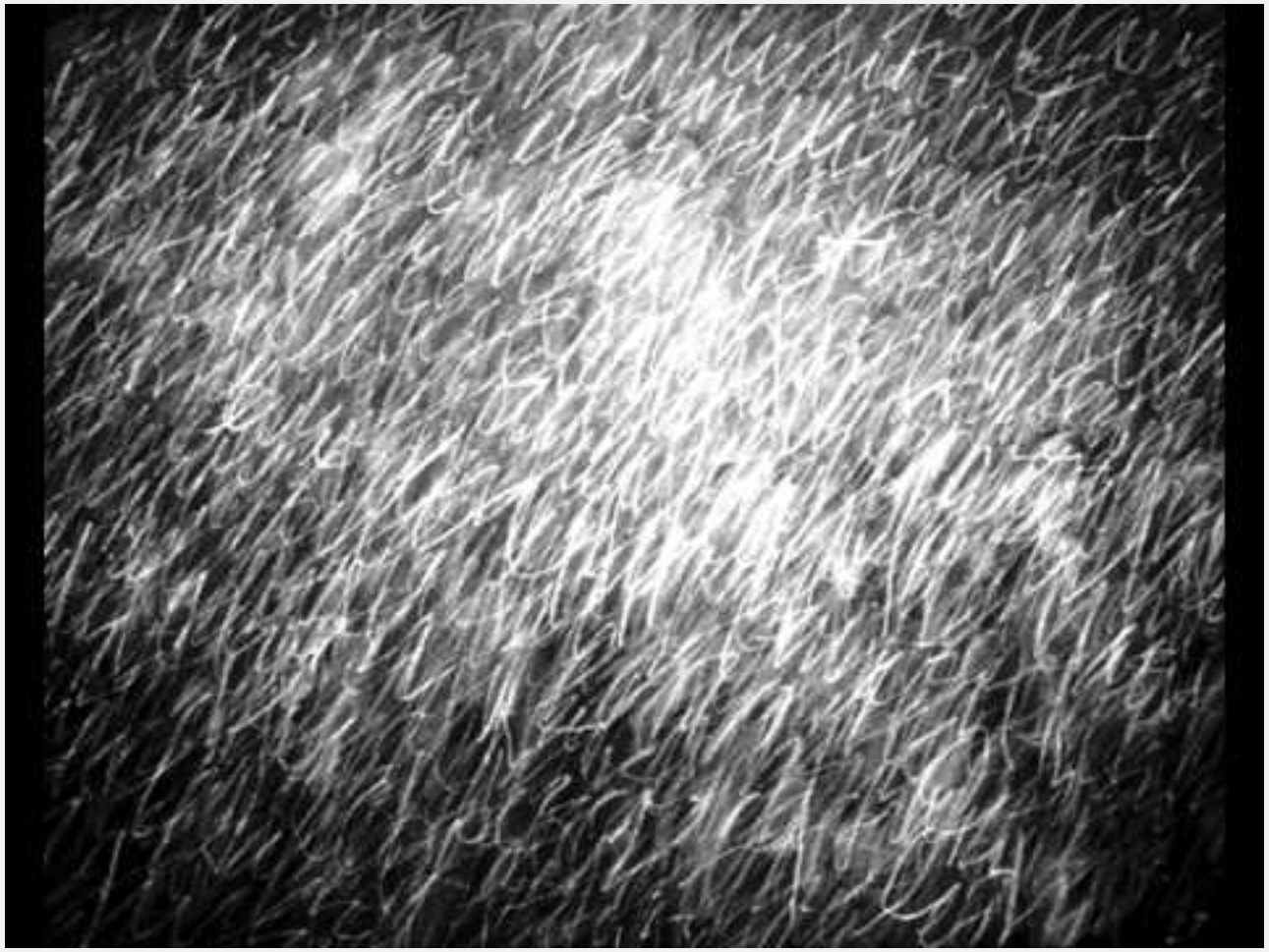
Early Cut and Paste

- A leading figure in electronic music, Stockhausen's early works in the 1950s involved manipulating tape.
- His pieces like *Gesang der Jünglinge* (1955-56) and *Kontakte* (1958-60) used tape techniques in combination with electronic sounds



Early Cut and Paste

- Although John Cage is more widely known for his chance-based and avant-garde works, his use of tape in pieces like Williams Mix (1952) reflects an early exploration of the potential of splicing and editing pre-recorded sounds.



Early Cut and Paste

- Co-founder of the BBC Radiophonic Workshop, Daphne Oram was one of the first composers in the UK to experiment with tape and sound manipulation. Her work in the late 1950s and 1960s contributed to the development of electronic music.
- Her technique of drawing sound waves onto film to control pitch and timbre, known as *Oramics*, built on early tape manipulation practices.



Early Cut and Paste

- Also a member of the BBC Radiophonic Workshop, Delia Derbyshire used tape-splicing techniques to create complex sound collages, layering tape loops, cutting and reassembling tape to achieve precise results.



Into Popular Culture

- In the 1960s, the Mellotron was an early instrument that used tape loops to simulate orchestral sounds.
- Although it was not strictly a sampling device in the modern sense, its ability to replay pre-recorded sounds via keyboard was an important step toward the sampling culture of the 1970s and beyond.
- It was famously used by bands like The Beatles (e.g. "Strawberry Fields Forever") and King Crimson.



Into Popular Culture

- In the late 1970s and early 1980s, hip-hop DJs such as Kool Herc, Grandmaster Flash, and Afrika Bambaataa began to use turntables in innovative ways, manipulating vinyl records as a form of live "sampling"
- This was an early form of musical sampling that didn’t yet rely on digital tools, but instead used physical media.



Into Popular Culture

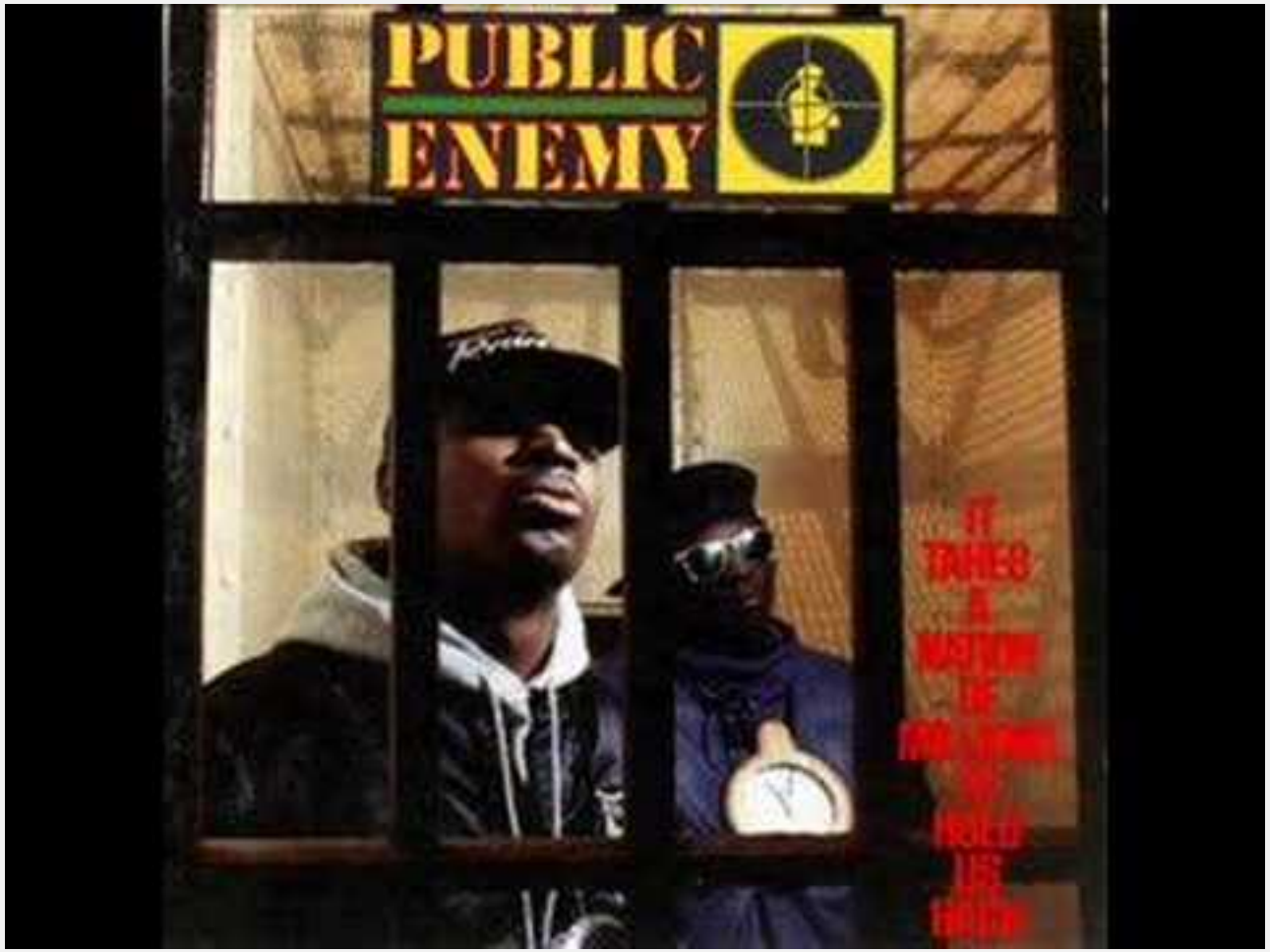
- In the late 1970s, the Fairlight CMI became one of the first commercially available digital sampling instruments. Pioneers like Peter Gabriel, Kate Bush, and Jean-Michel Jarre used the Fairlight to sample real-world sounds and manipulate them digitally.
- The Fairlight played a key role in shaping the sound of 1980s pop and experimental music, with songs like Gabriel's "Shock the Monkey" or Bush's Hounds of Love heavily relying on sampling.



9,25 for Kate Bush

Into Popular Culture

- The rise of the E-MU SP-1200 and Akai MPC samplers in the 1980s revolutionised hip-hop music. Producers like Marley Marl, DJ Premier, and The Bomb Squad (who produced Public Enemy) pioneered the art of sampling short sound bites from other records to create new, complex sound collages.
- Public Enemy's *It Takes a Nation of Millions to Hold Us Back* (1988) is often cited as a landmark in sampling, blending hundreds of tiny sound fragments into a cohesive whole.



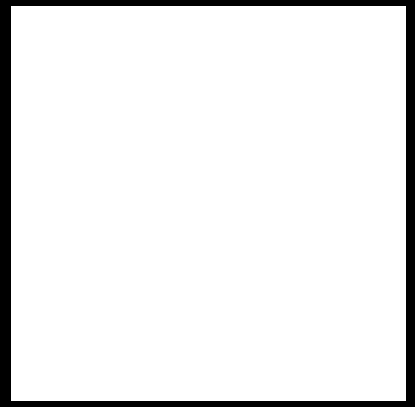
Plunderphonics

- John Oswald's 1989 album Plunderphonics was a manifesto of sorts, built entirely from fragments of existing popular music.
- For example, in his track "Dab," he transformed Michael Jackson’s "Bad" by slowing it down and restructuring it, creating a warped and often surreal version of the original.
- The project was quickly targeted by legal action from record companies due to copyright infringement, but it laid the groundwork for the concept of Plunderphonics as a distinct artistic practice.



Mashup

- DJ Shadow, The Avalanches, DJ Yoda, DJ Food are all good examples of more recent artist who have layered (sometimes) thousands of samples together into cohesive compositions
- They often recombined popular music into new pieces
- Special shout to DJ Food - Raiding the 20th Century (Words & Music Expansion) feat. Strictly Kev, Paul Morley & a Cast of Thousands...
  - https://soundcloud.com/chimpski/dj-food-raiding-the-20th?in=lidoboscoverde/sets/lido-bv



Cut and Paste in Python

